Fields and Waves I

Lecture 3

Sine Waves on Transmission Lines II

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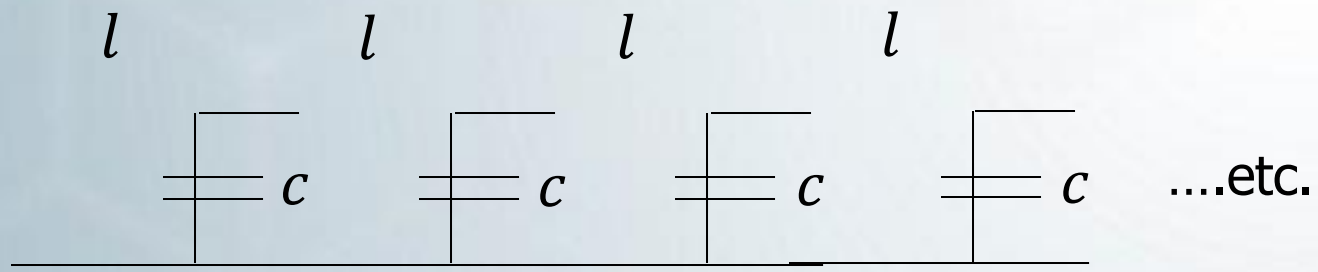
Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

Outline

- Reflections on Transmission Lines
- Reflection and Standing Waves
- Generalized Impedance
- Input Impedance

Review

Model the full length of lossless transmission line as:



Using KVL and KCL, we derived two differential equations in terms of voltage and current on the transmission line

$$\frac{\partial}{\partial z}V(z,t) = -l\frac{\partial}{\partial t}I(z,t)$$

$$\frac{\partial}{\partial z}I(z,t) = -c\frac{\partial}{\partial t}V(z,t)$$

$$\rightarrow \quad \frac{\partial^2}{\partial z^2}V(z,t) = lc\frac{\partial^2}{\partial t^2}V(z,t)$$

This is a wave equation!

Review

- The solutions take the form of traveling waves:

$$A\cos(\omega t \mp \beta z)$$

- Another way to write this equation is:

$$A\cos(\omega t \mp \beta z)$$

- T is the time period, λ is the wavelength

- Therefore the first term in the argument tells us how rapidly the wave changes in *time*. The second tells us how rapidly it changes in *space*.

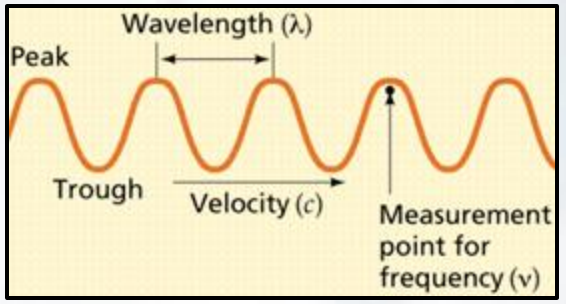


Figure from http://www.emc.maricopa.edu/

Phase constant

$$\beta = \frac{\omega}{u} = \omega\sqrt{lc} = \frac{2\pi}{\lambda}$$

Frequency

$$\omega = 2\pi f = \frac{2\pi}{T}$$

Wavelength

$$\lambda = \frac{2\pi}{\beta} = \frac{u}{f}$$

Velocity

$$u = \frac{1}{\sqrt{lc}}$$

Review

In the phasor domain, the solutions to the Telegrapher's equations take the form:

$$\tilde{V}(z) = V_o^+ e^{-uz} + V_o^- e^{uz}$$

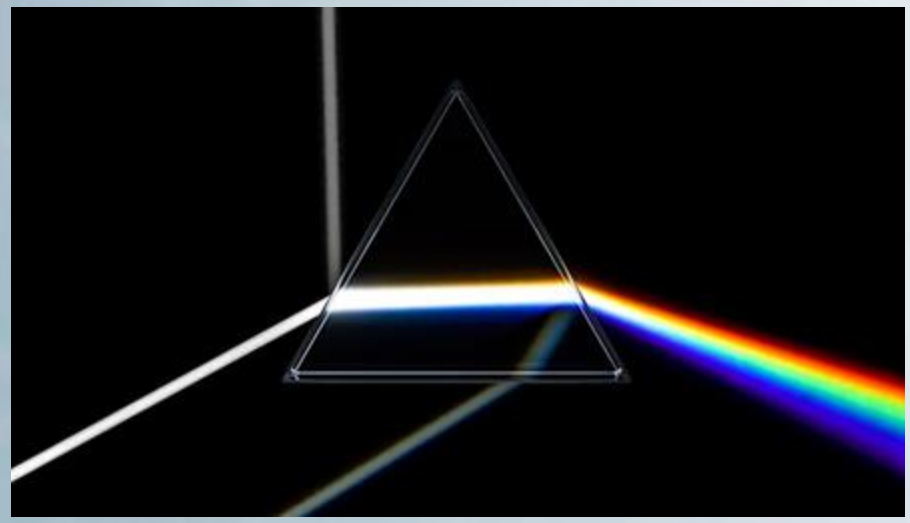
$$\tilde{I}(z) = \frac{u}{j\omega l}\left(V_o^+ e^{-uz} - V_o^- e^{uz}\right)$$

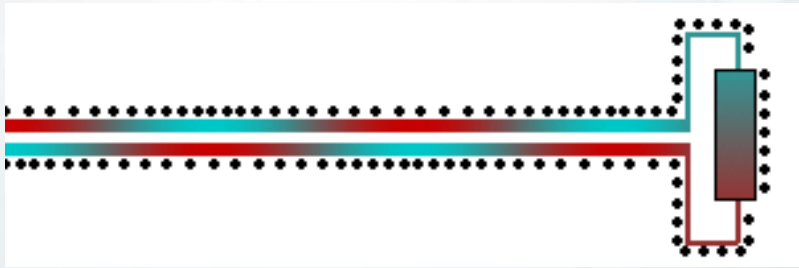
… such that the voltage/current ratio for a given traveling wave is a constant. (This is the expression for **characteristic impedance** - The impedance that a signal "sees" at the input of transmission line when the load is far away (when there is no signal on it)

$$Z_0 = \frac{V_0^+}{I_0^+} = \frac{-V_0^-}{I_0^-} = \frac{j\omega l}{u} = \frac{j\omega l}{\sqrt{(j\omega l)(j\omega c)}} = \sqrt{\frac{l}{c}}$$

Wave Reflection

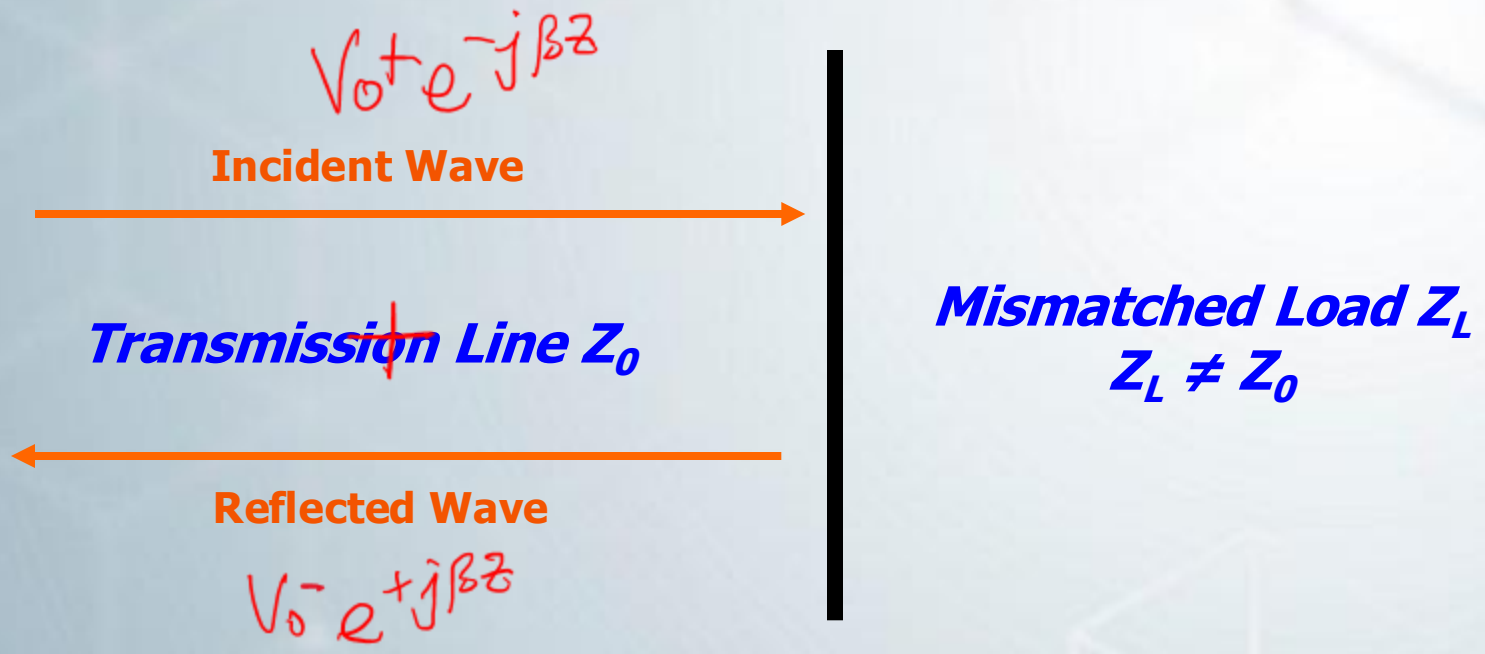
Wave Reflection





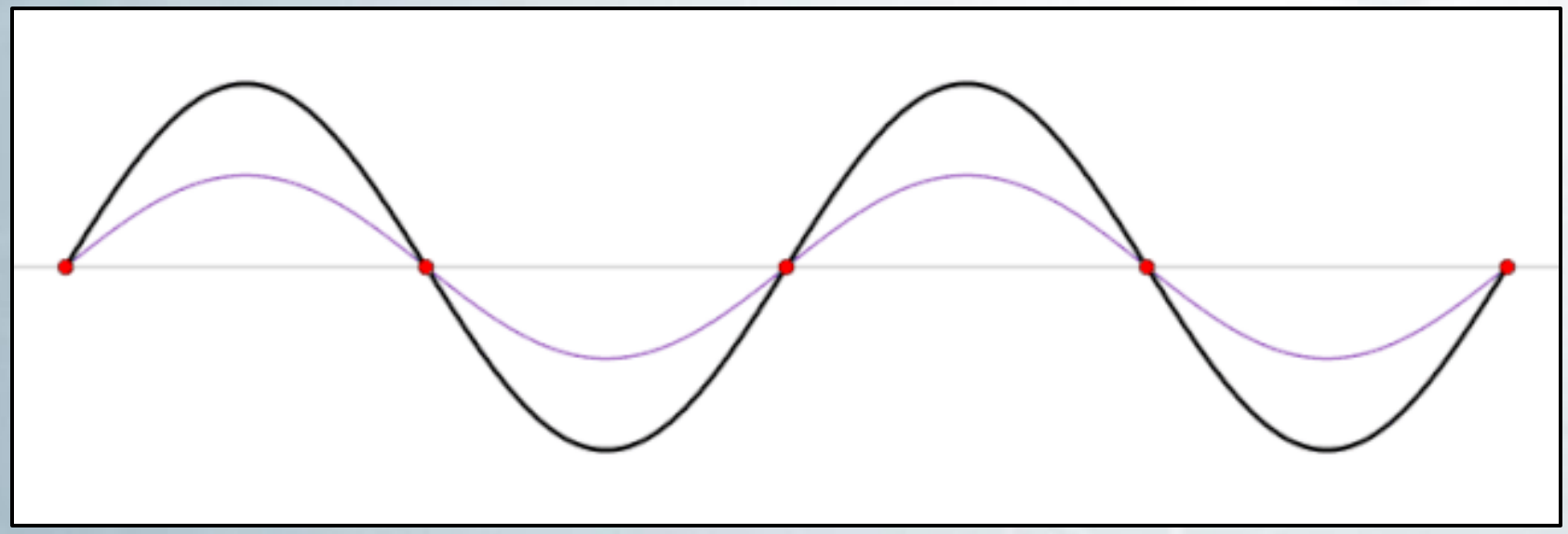
- When light enters a prism, some light will be transmitted and some will be reflected. This is because the **impedance** of the prism to light is different to that of air.
- Likewise, a signal that travels down a transmission line will reflect at the load if the **characteristic impedance of the line** and the **impedance of the load** are **not the same**.

Wave Reflection



These two waves interfere with each other and can produce a **standing wave**.

Wave Reflection



These two waves interfere with each other and can produce a **standing wave**.

Wave Reflection

- With a reflected wave term, the total voltage phasor on the line is:

$$V = V_0^+ e^{-j\cdot\beta\cdot z} + V_0^- e^{+j\cdot\beta\cdot z}$$

$cos(\omega\cdot t - \beta\cdot z)$ *TIME DOMAIN* $cos(\omega\cdot t + \beta\cdot z)$

- The forward- and backward-moving waves will interfere on the line:

V_{max} occurs when forward and backward waves are in phase

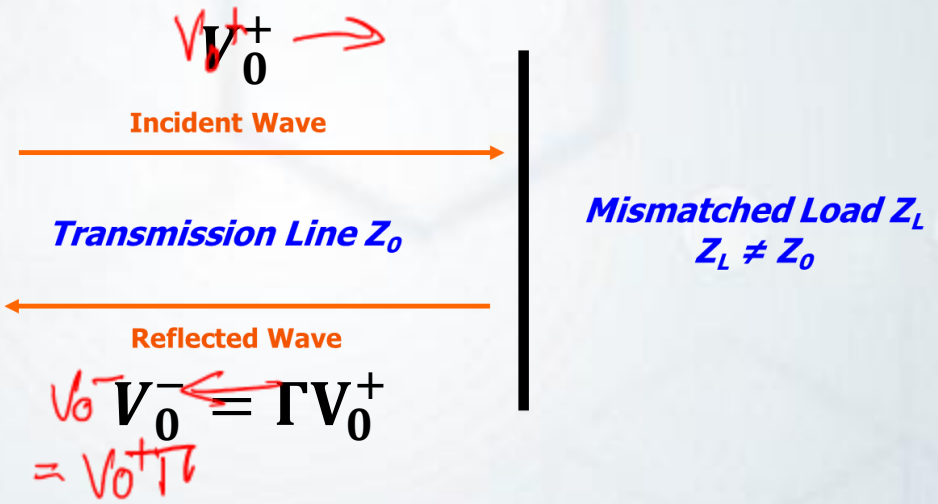
⇨ ***CONSTRUCTIVE INTERFERENCE***

V_{min} occurs when forward and backward waves are out of phase

⇨ ***DESTRUCTIVE INTERFERENCE***

Wave Reflection

- So, how much gets reflected?



- Define the **Reflection Coefficient** Γ:

$$V_0^+ \Gamma = V_0^-$$

- This can be done in terms of either current or voltage:

$$\Gamma = \frac{V_0^-}{V_0^+} = \frac{-I_0^-}{I_0^+}$$

Wave Reflection

- We can define load impedance as follows:

$$Z_L = \frac{\tilde{V}_L}{\tilde{I}_L}$$

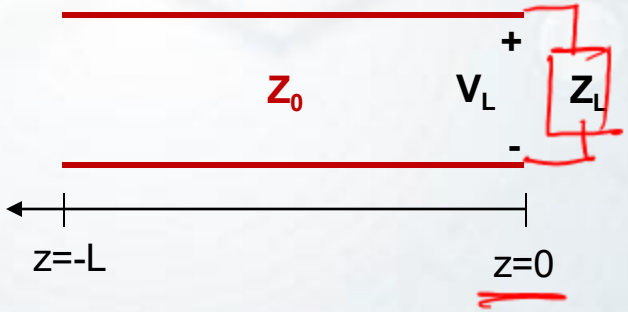
- Meanwhile we previously said (general solution to Telegrapher's Equations):

$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{+j\beta z}$$

$$\tilde{I}(z) = I_0^+ e^{-j\beta z} - I_0^- e^{+j\beta z}$$

Wave Reflection

$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{+j\beta z}$$
$$\tilde{I}(z) = I_0^+ e^{-j\beta z} - I_0^- e^{+j\beta z}$$



- When z=0 (i.e. we are at the load), we get:

$$\tilde{V}_L = \tilde{V}(z=0) = V_0^+ + V_0^-$$

$$\tilde{I}_L = \tilde{I}(z=0) = \frac{V_0^+}{Z_0} - \frac{V_0^-}{Z_0}$$

- Calculating Z_L:

$$\frac{\tilde{V}_L}{\tilde{I}_L} = Z_L = \left(\frac{V_0^+ + V_0^-}{V_0^+ - V_0^-}\right) Z_0$$

- Doing some algebra…

$$V_o^- = \left(\frac{Z_L - Z_0}{Z_L + Z_0}\right) V_o^+ = \Gamma V_{0+}$$

gamma Γ

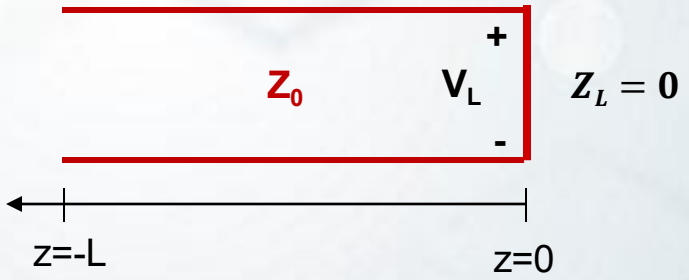
Reflection Coefficient

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Wave Reflection

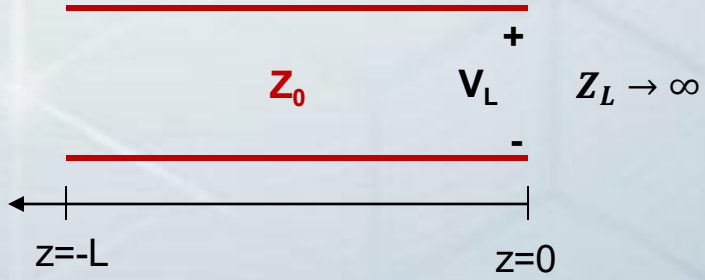
- At short circuit, load impedance is zero.

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{0 - Z_0}{0 + Z_0} = -1$$



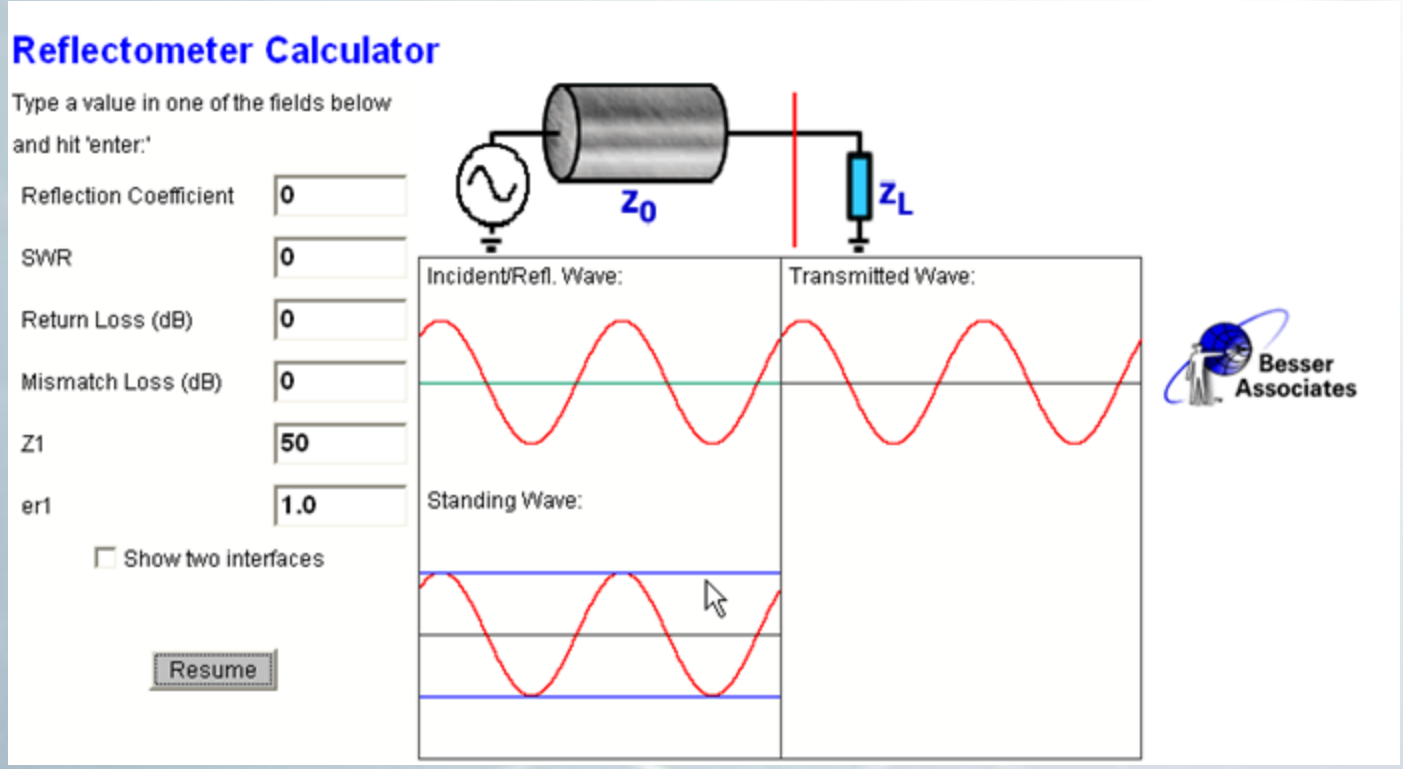
- At open circuit, load impedance is *very big.*

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{very\ big - Z_0}{very\ big + Z_0} = 1$$



Reflection and Standing Waves

Reflection and Standing Waves



Reflection and Standing Waves

Deriving a Standing Wave Expression

- Voltage general wave solution (as a phasor):

$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{+j\beta z}$$

- Using the reflection coefficient:

$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + \Gamma V_0^+ e^{+j\beta z}$$

Reflection and Standing Waves

Short Circuit Load Case

- For $Z_L=0$, we have

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{0 - Z_0}{0 + Z_0} = -1$$

- Plugging in $\Gamma = 1$ and converting to the sinusoidal form:

$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + \Gamma V_0^+ e^{+j\beta z} = V_0^+ (e^{-j\beta z} - e^{+j\beta z})$$

$$e^{+j\beta z} = \cos(\beta z) + j\sin(\beta z)$$

$$e^{-j\beta z} = \cos(\beta z) - j\sin(\beta z)$$

$$\Rightarrow \quad \tilde{V}(z) = -j2V_0^+ \sin(\beta z)$$

total voltage phasor on t-line in steady state for a short circuit

Reflection and Standing Waves

Short Circuit Load Case

- Convert to space-time form

$$\tilde{V}(z,t) = Re\left\{\tilde{V}e^{j\omega t}\right\} = Re\left\{-j2V_0^+ \sin(\beta z)e^{j\omega t}\right\}$$



$$\tilde{V}(z,t) = 2V_0^+ \sin(\beta z)\sin(\omega t)$$

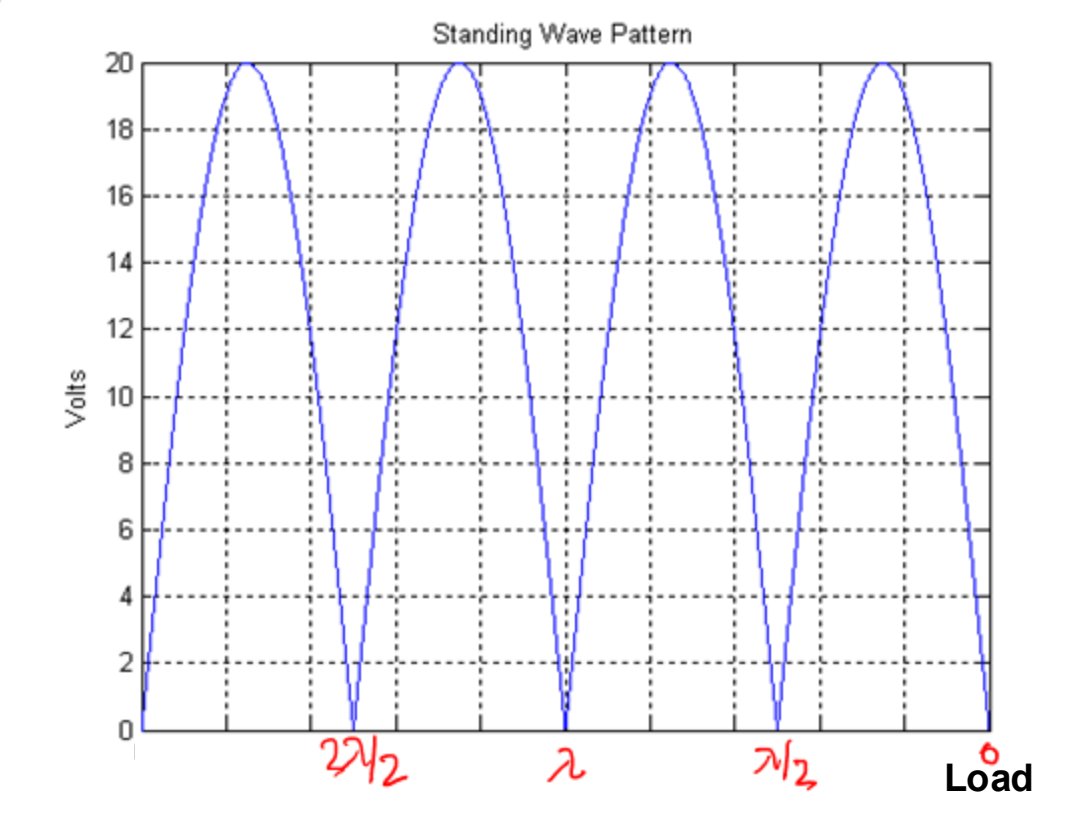
total voltage on t-line in time domain

- Is this is a standing wave? How do we know?

yes!

Reflection and Standing Waves

Short Circuit Load Case



We can characterize the extent to which a wave has non-traveling-wave properties via its **Standing Wave Ratio**

$$\textbf{SWR} = \frac{V_{max}}{V_{min}} = \frac{1+|\Gamma_L|}{1-|\Gamma_L|}$$

This is a voltage **standing wave plot** - a plot of the **maximum magnitude** of the sum of the forward and backward moving waves at each position along the transmission line : $\max\left(|\tilde{V}(z)|\right)$

Reflection and Standing Waves

Short Circuit Load Case

- What are the voltage maxima and minima of the standing wave pattern?

$$V(z,t) = 2V^{+}\sin(\beta z)\sin(\omega t)$$

- Voltage minima: $\beta z = 0, \pi, 2\pi$, etc. (where sin = 0)

  Since $\beta = 2\pi/\lambda$,

  $z = 0, \lambda/2, \lambda, 3\lambda/2$, etc.

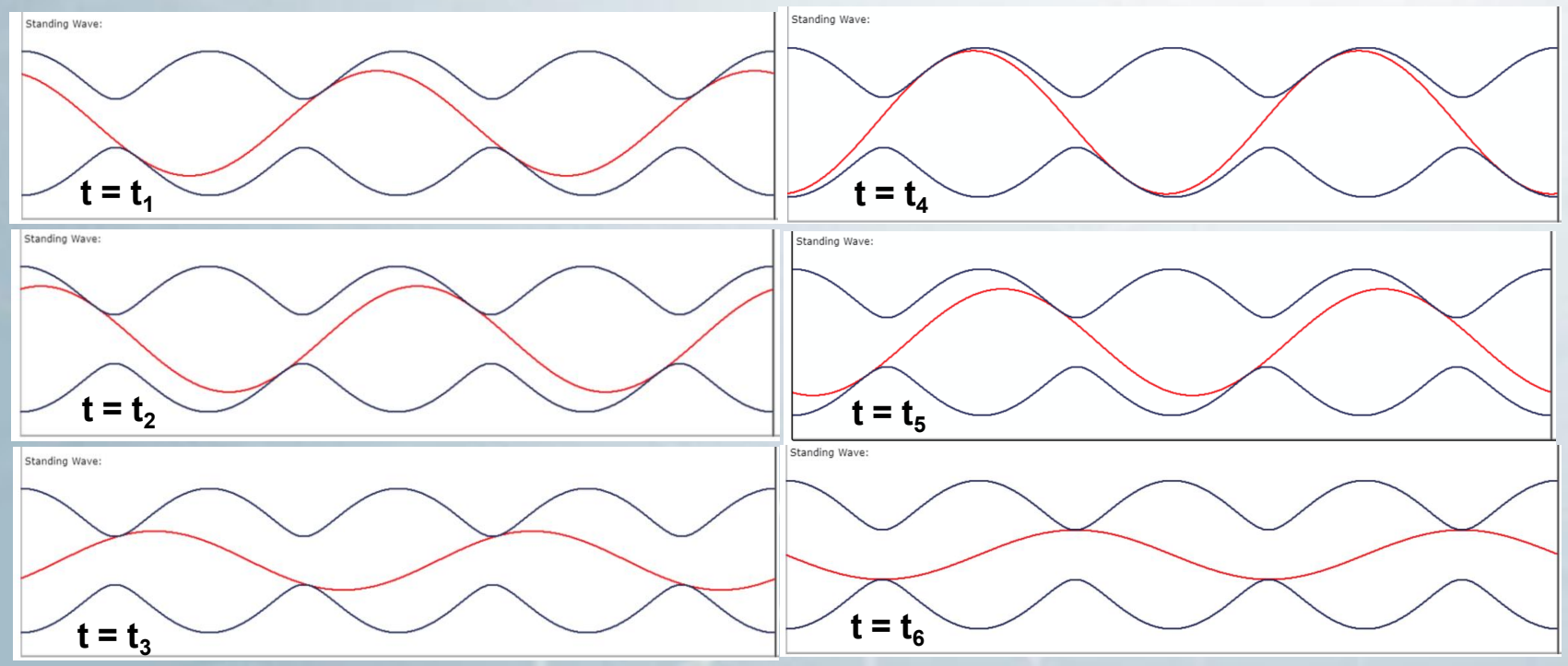
- Voltage maxima: $\beta z = \pi/2, 3\pi/2$, etc. (where sin = 1)

  $z = \lambda/4, 3\lambda/4, 5\lambda/4$, etc.

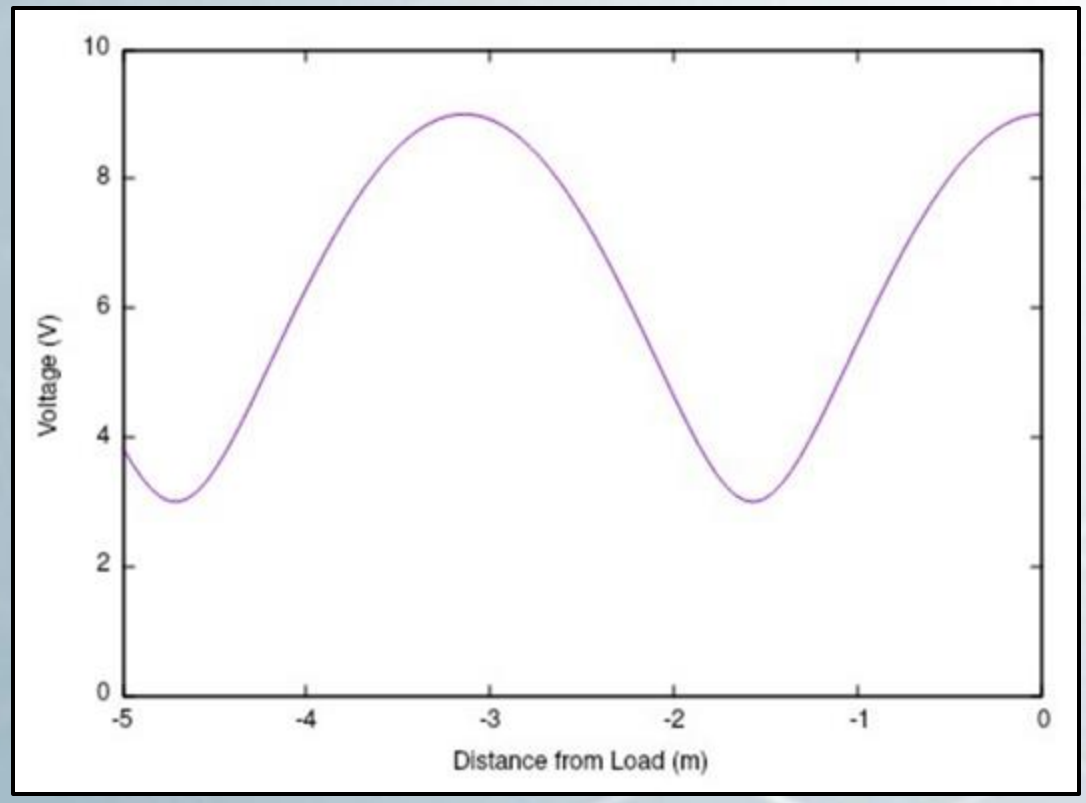
In general, standing wave patterns will repeat every half wavelength of the input signal, and every quarter wavelength they will alternate between maxima and minima.

Reflection and Standing Waves

- The standing wave pattern is the **envelope** of the total voltage or current function - a graph of its maximum magnitude at each point on the z-axis (over time)



Reflection and Standing Waves



Z_0 = 50Ω

- First, what is the SWR?

$$SWR = \frac{V_{max}}{V_{min}} = \frac{9V}{3V} = 3$$

Let's say that you're given a standing wave graph like the one above **and Z_0 of the line**, but no other information. How do you extract key wave / line parameters?

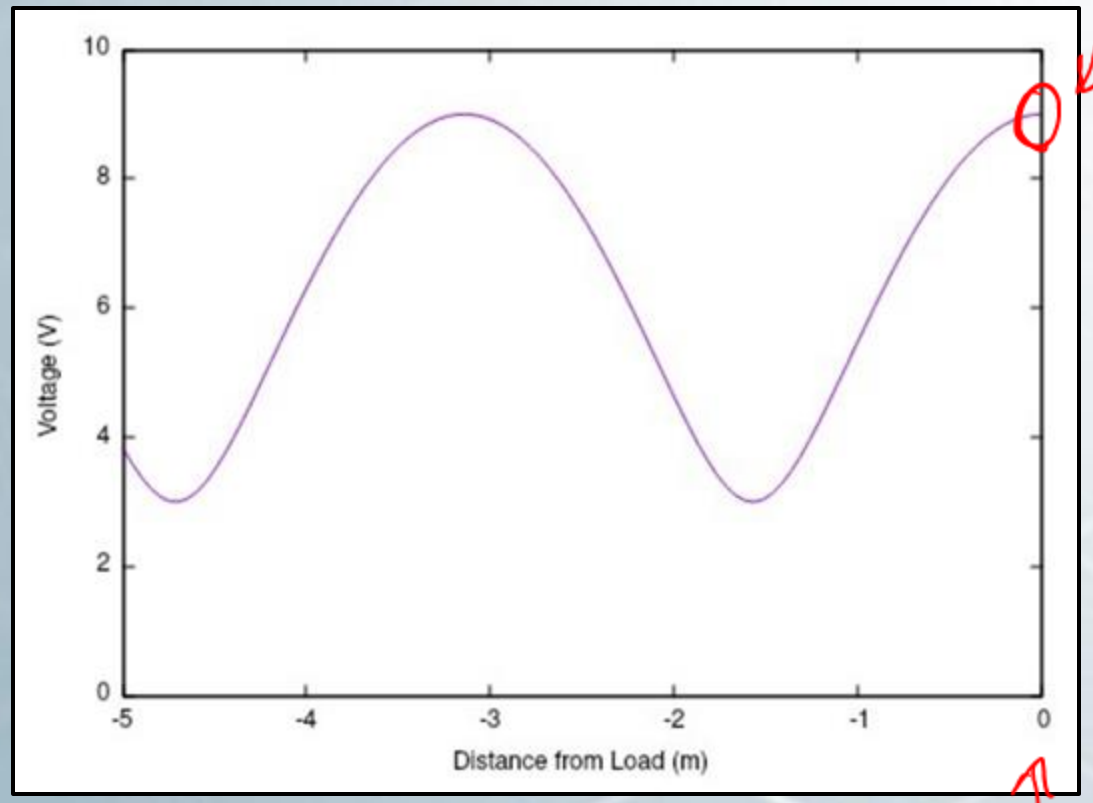
Reflection and Standing Waves

- Next, what is the reflection coefficient? **$Z_0 = 50\ \Omega$**

$$SWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} = 3 \rightarrow 1 + |\Gamma| = 3 - 3|\Gamma|$$

$$\rightarrow 4|\Gamma| = 2 \rightarrow |\Gamma| = 0.5$$

Reflection and Standing Waves



maximum

load

R_s + V_s R_L V_L −

$Z_0 = 50\Omega$

$$\Gamma = \pm\frac{1}{2}$$

To figure out the sign of Γ,

look at whether the load end of the graph is a **maximum or a minimum**.

Reflection and Standing Waves

- Let's look at the expression for voltage at the load again to help us out:

$$V(z) = V_0^+ e^{-j\beta z} + \Gamma_L V_0^+ e^{+j\beta z}$$

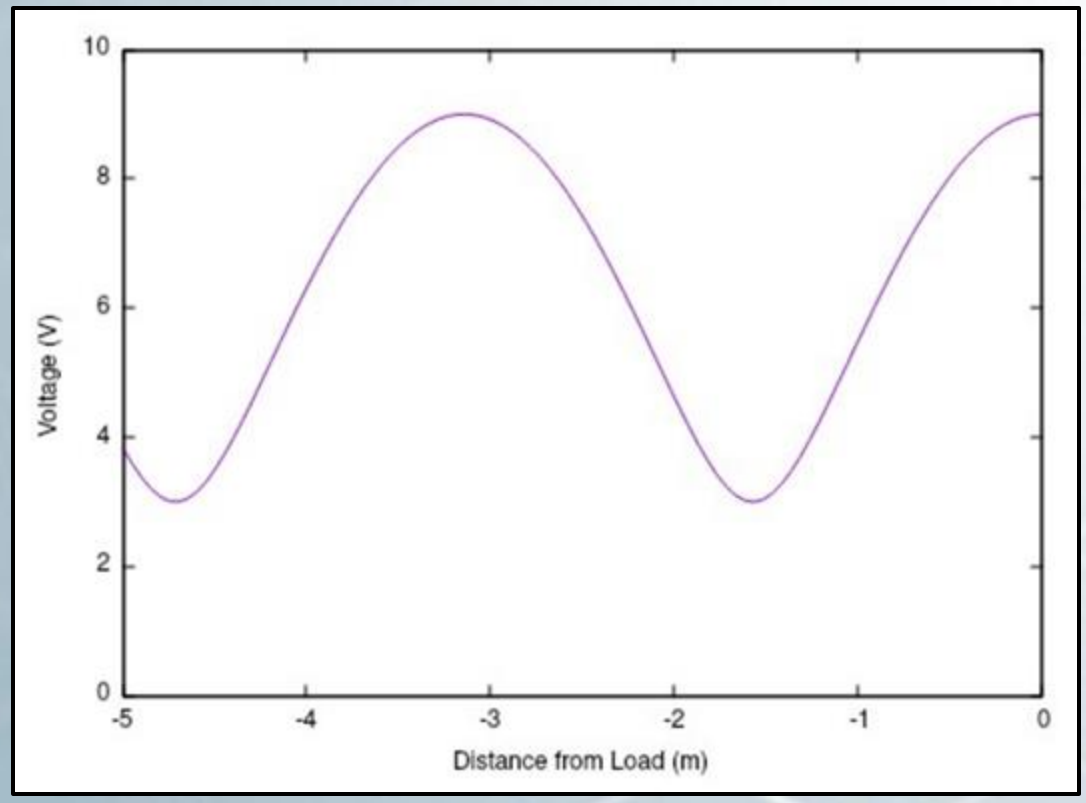
At the load,

$$V(0) = V_0^+ + \Gamma_L V_0^+$$

Therefore:

- If the load is not matched to the line, **and it's a resistive load,** there will be either a voltage maximum or voltage minimum at the load.
- If the load has a **voltage maximum** (meaning $V(0) > V_0^+$), then Γ is **positive**
- If the load has a **voltage minimum** (meaning $V(0) < V_0^+$), then Γ is **negative**

Reflection and Standing Waves



$Z_0 = 50\Omega$

$$\Gamma = \pm\frac{1}{2}$$

So what is Γ for this line?

$\Gamma = +0.5$

and $Z_L > Z_0$

Reflection and Standing Waves

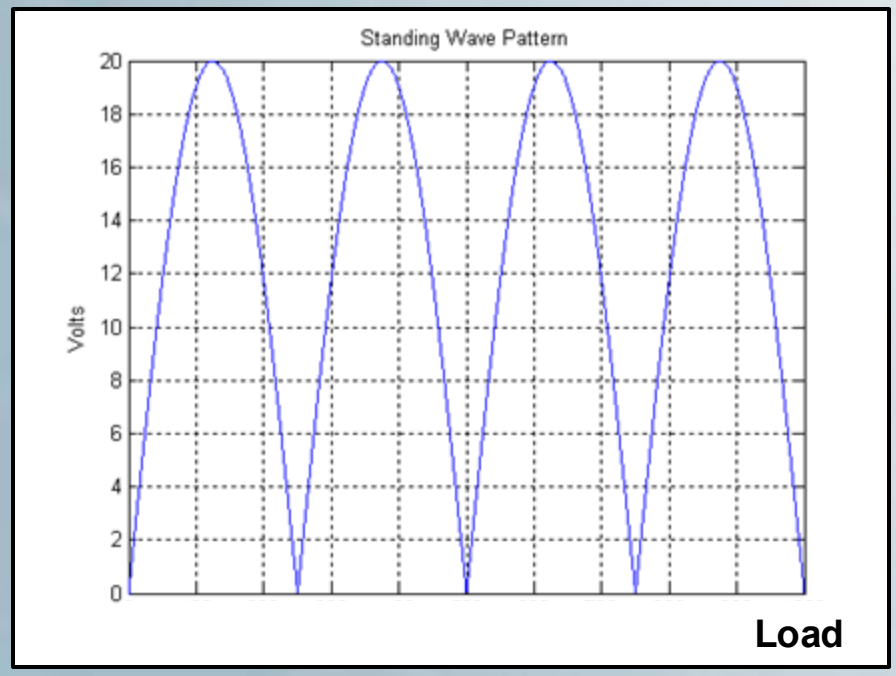
- We can now calculate the load impedance: $Z_0 = 50\Omega$

$$\Gamma = 0.5 = \frac{Z_L - Z_0}{Z_L + Z_0} \quad (Z_0 = 50\Omega) \rightarrow 0.5 Z_L + 25\Omega = Z_L - 50\Omega$$

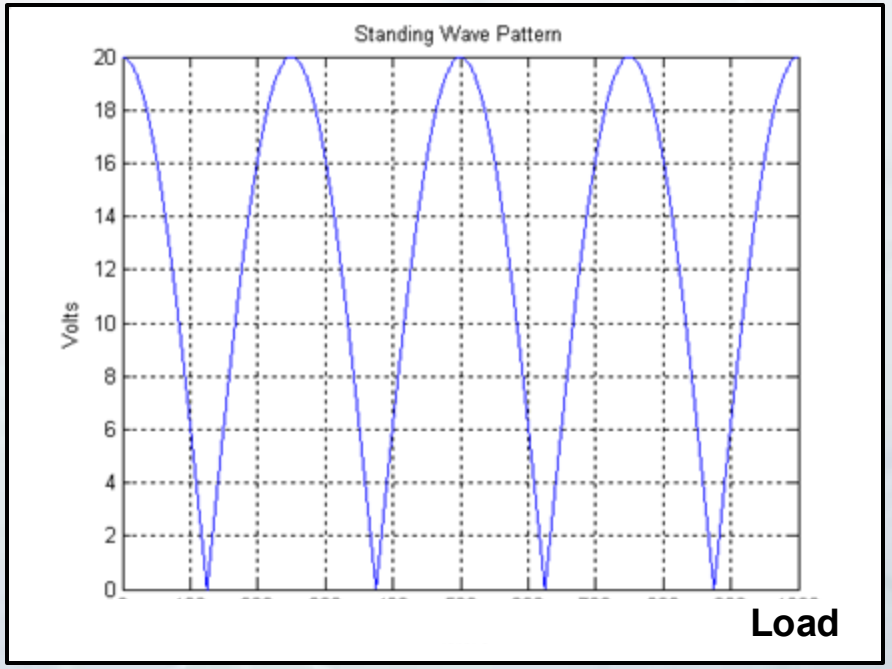
$$50\Omega \Rightarrow 75\Omega = 0.5 Z_L \rightarrow \underline{Z_L = 150\,\Omega}$$

Reflection and Standing Waves

Short-Circuit Load Standing Wave



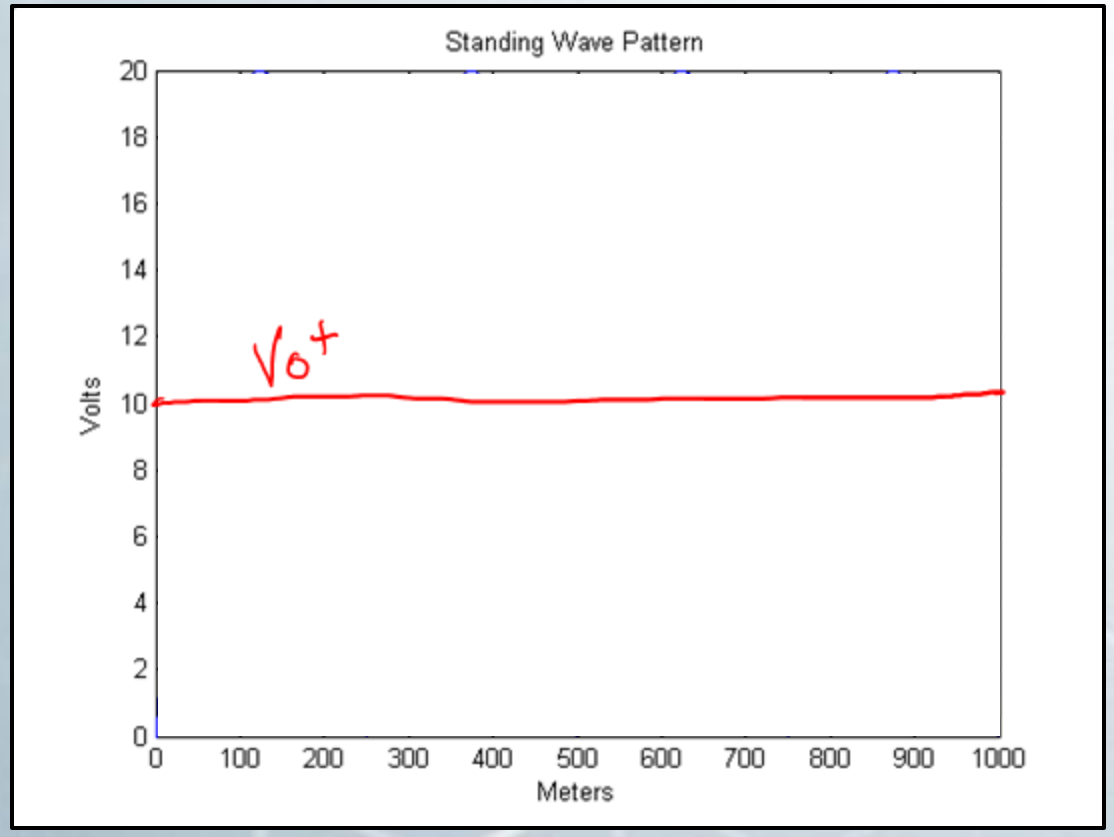
Open-Circuit Load Standing Wave



- Note that standing wave patterns of pure standing waves have an SWR of infinity since $V_{min} = 0$.
- The standing wave patterns are identical except for a **quarter-wavelength phase shift.**

Reflection and Standing Waves

- What will the standing wave pattern look like on a matched transmission line $Z_0 = Z_L$?



$$SWR = \frac{10V}{10V} = 1$$

$1 \leq SWR < \infty$

↑ matched ↑ full reflection

Reflection and Standing Waves

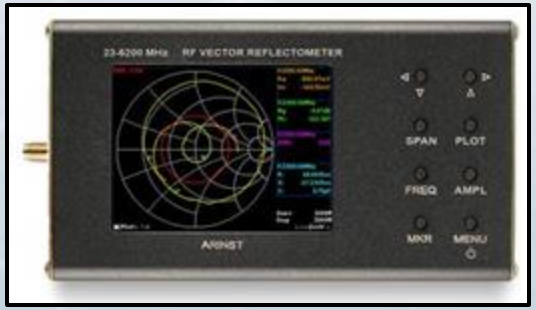
- In most practical applications, we try to **minimize standing wave ratio** since this represents power reflected away from the load
- If the voltage maxima created by a standing wave pattern exceed the rated voltage of a transmission line, it can also cause the transmission line to fail.
- The SWR of real lines can be measured using several different devices and methods

Directional Coupler



krytar.com

Spectrum Analyzer



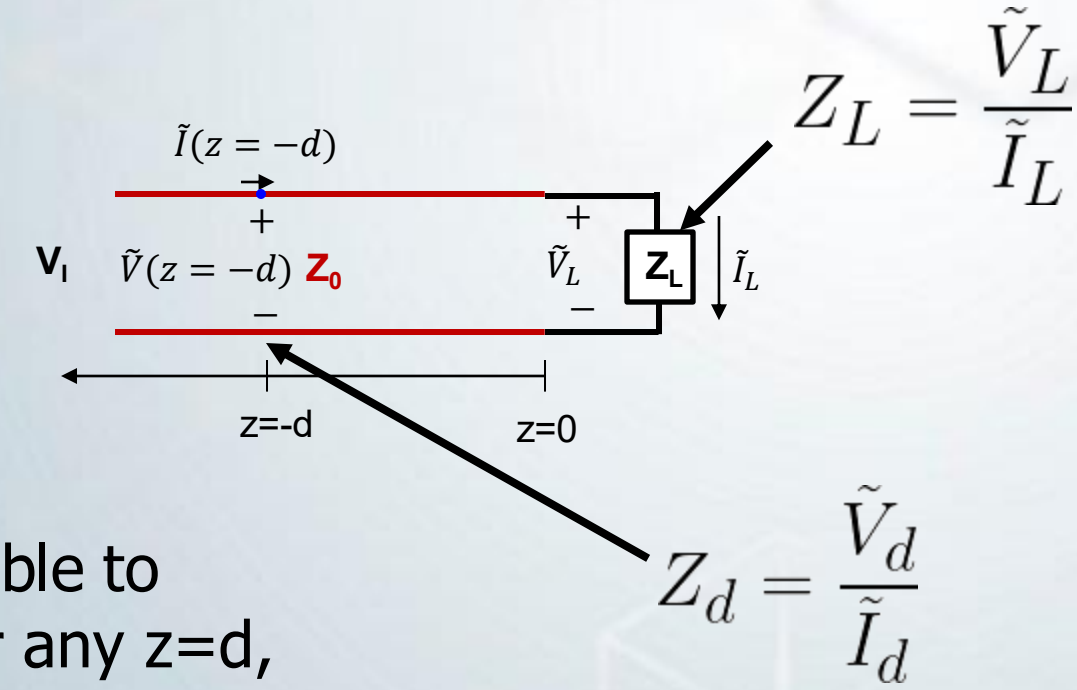
aliexpress.com

Form into group of up to 4 and do Lecture 03 Exercise 1 on Gradescope.

Generalized Impedance (Wave Impedance)

Generalized Impedance (Wave Impedance)

- We already saw that:



- But we should also be able to state more generally for any z=d, where "d" is some distance away from the load end of the t-line:

- It's not clear yet what an impedance like this actually represents.

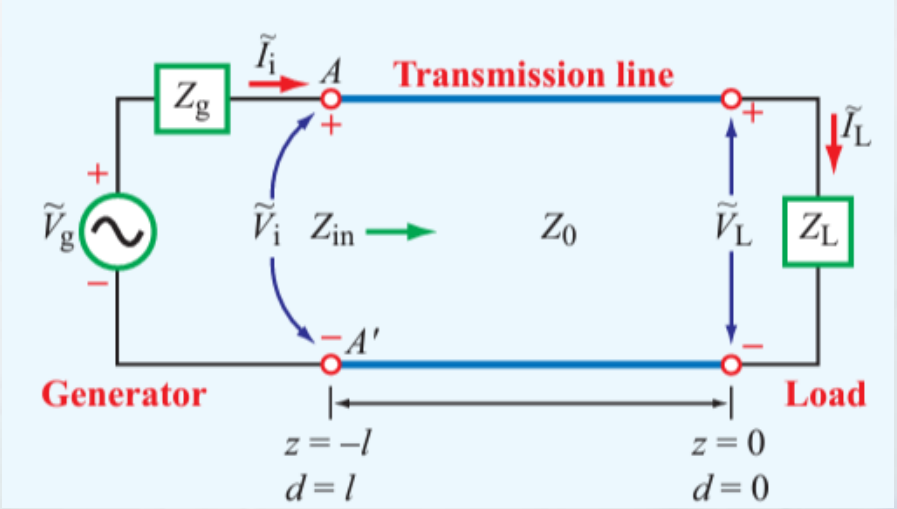
Generalized Impedance (Wave Impedance)

$$\frac{\tilde{V}_d}{\tilde{I}_d} = \frac{V_0^+[e^{j\beta d} + \Gamma e^{-j\beta d}]}{V_0^+[e^{j\beta d} - \Gamma e^{-j\beta d}]} Z_0$$

refl. coeff. at load

$$= Z_0 \frac{1 + \Gamma e^{-j2\beta d}}{1 - \Gamma e^{-j2\beta d}}$$

$$= Z_0 \frac{1 + \Gamma_d}{1 - \Gamma_d}$$

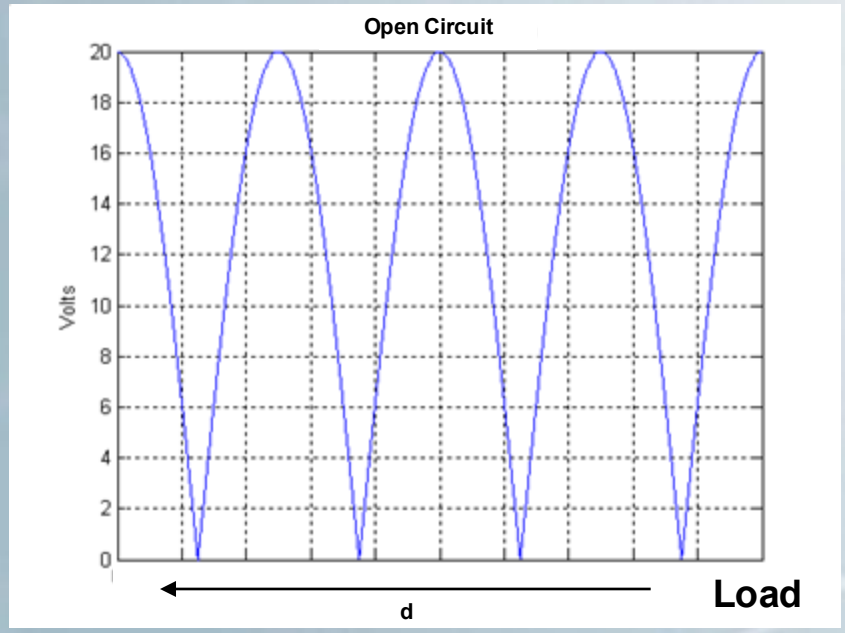


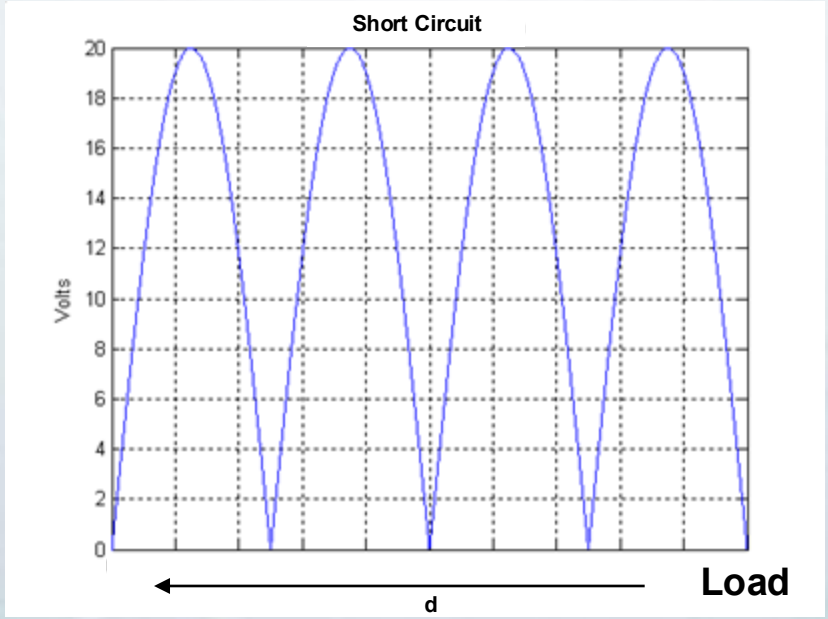
Phase-shifted voltage reflection coefficient

$$\Gamma_d = \Gamma(z) = \Gamma e^{-2j\beta d}$$

Generalized Impedance (Wave Impedance)

- Looking at the voltage standing wave pattern on the line, what trends do we see vs. distance from the load?





Generalized Impedance (Wave Impedance)

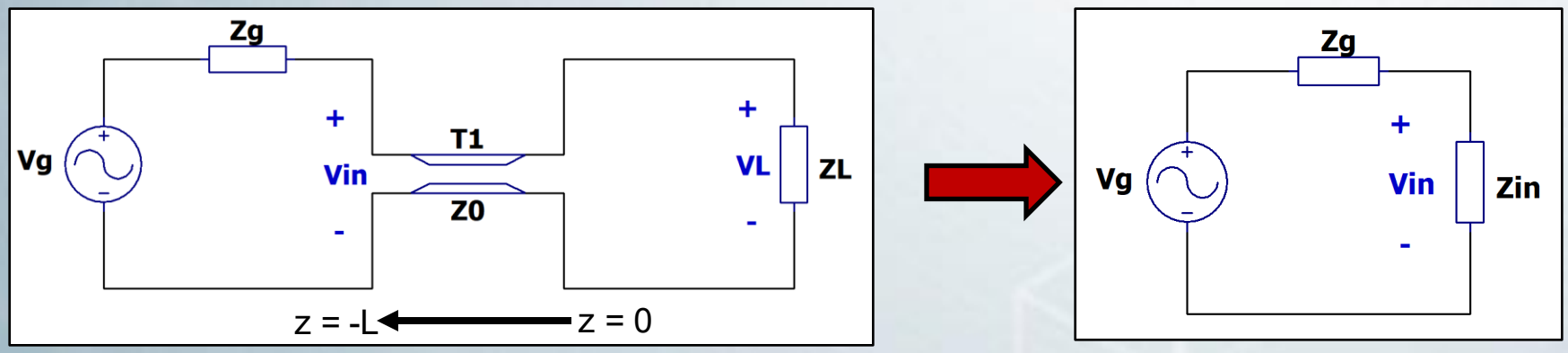
$$\Gamma_d = \Gamma(z) = \Gamma e^{-2j\beta d}$$

- We see that going backward a quarter wavelength from an open circuit load produces the same voltage pattern you would see if there were a short circuit load located just to the right of that point.
- In other words, as you move backward from the load, it begins to "look like" a different load.
- The expression for Γ(z), the phase-shifted reflection coefficient, takes into account the load as well as the distance from it.
- If we travel all the way back to the beginning of the transmission line, we can figure out what load the source circuit sees when we connect a t-line to it. This is called the **input impedance** of the transmission line.

Input Impedance

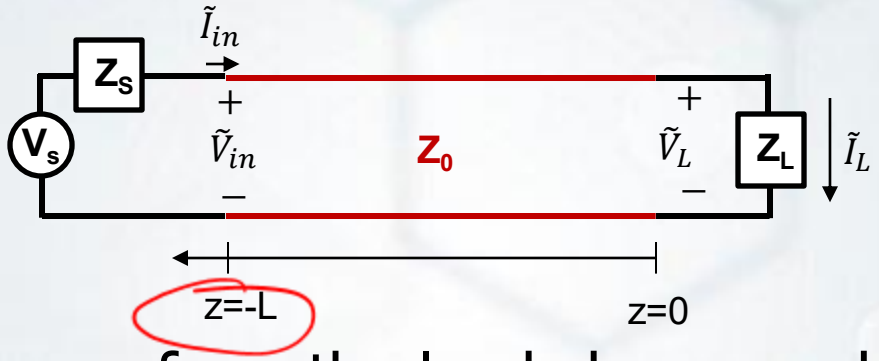
Input Impedance

- Consider a transmission line of length L.
- If we know $Z(z = -L)$ (also known as Z_{in}), we can replace our t-line and load with a single element, Z_{in}:



We can replace the entire transmission line and the load impedance in a circuit with just Z_{in}

Input Impedance



- If the distance on the transmission line away from the load changes what the load impedance "looks like", then what does the impedance looking into the **input** of the transmission line (Z_{in}),

$$Z_{in} = \frac{V_{in}}{I_{in}}$$ look like?

- To figure this out, we can start by using our expression for the phase-shifted reflection coefficient.

$$\Gamma(z) = \frac{V^- \cdot e^{+j\cdot\beta\cdot z}}{V^+ \cdot e^{-j\cdot\beta\cdot z}} = \frac{V^-}{V^+} \cdot e^{j\cdot 2\cdot\beta\cdot z}$$

we replaced z with −d in the previous slides

$$= \Gamma_L \cdot e^{j\cdot 2\cdot\beta\cdot z}$$ if z = 0 at the LOAD

Input Impedance

- Previously, we have seen:

$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{+j\beta z}$$

$$\rightarrow \tilde{V}(z = -d) = V_0^+ \left(e^{j\beta \mathrm{d}} + \Gamma_L e^{-j\beta \mathrm{d}}\right)$$

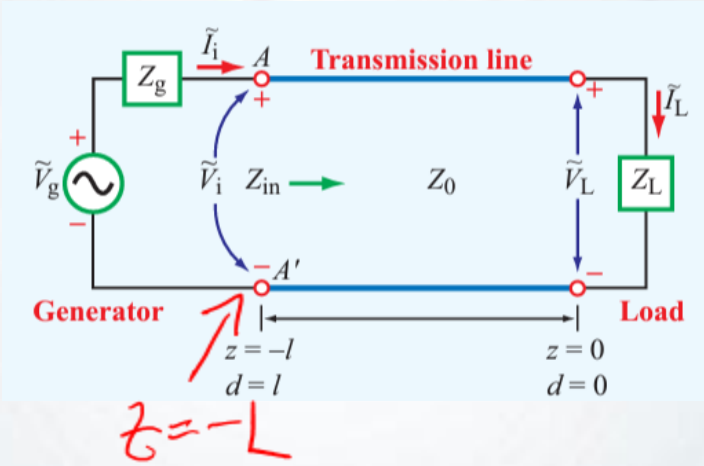
$$= V_0^+ e^{j\beta \mathrm{d}} (1 + \Gamma_d)$$

- What about I?

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-j\beta z} - \frac{V_0^-}{Z_0} e^{+j\beta z}$$

$$\rightarrow \tilde{I}(z = -d) = \frac{V_0^+}{Z_0} e^{j\beta \mathrm{d}} (1 - \Gamma_d)$$

Input Impedance



- From the ratio (the generalized impedance):

$$Z_d(z=-d) = Z_0 \frac{1+\Gamma_d(-d)}{1-\Gamma_d(-d)} \quad \text{where} \quad \Gamma_d(-d) = \Gamma_L e^{-j2\beta d}$$

- We are primarily interested in the z=-L value – this is the impedance at the **input to the transmission line**

$$\tilde{Z}_{in}(z=-L) = Z_0 \frac{1+\Gamma_d(-L)}{1-\Gamma_d(-L)} \quad \text{where} \quad \Gamma_d(-L) = \Gamma_L e^{-j2\beta L}$$

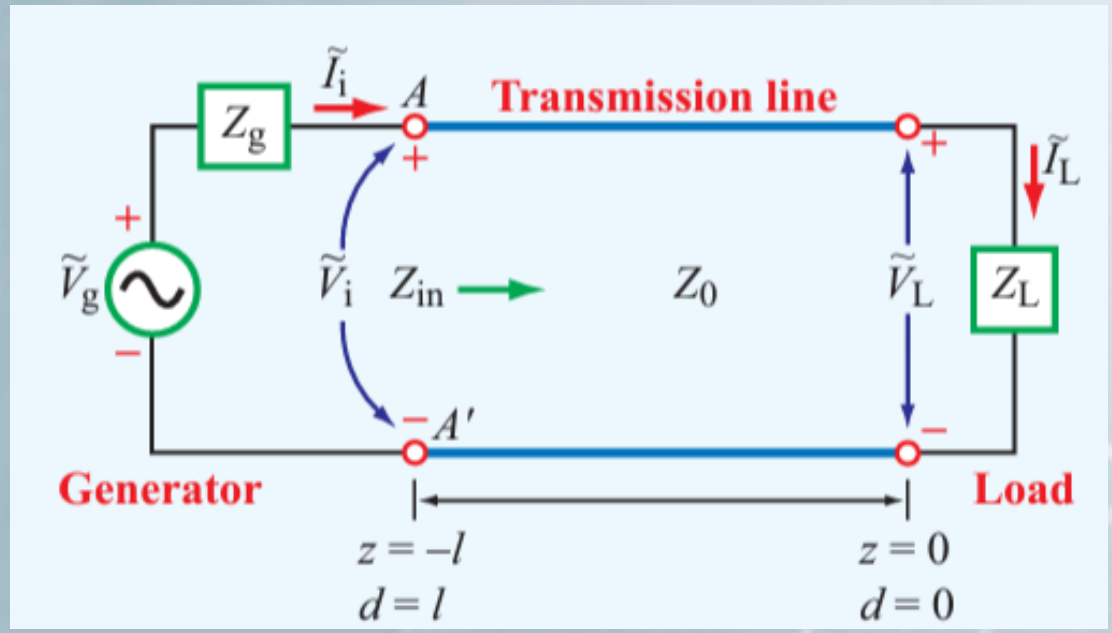
replace Γ_L w/ $\frac{Z_L - Z_0}{Z_L + Z_0}$

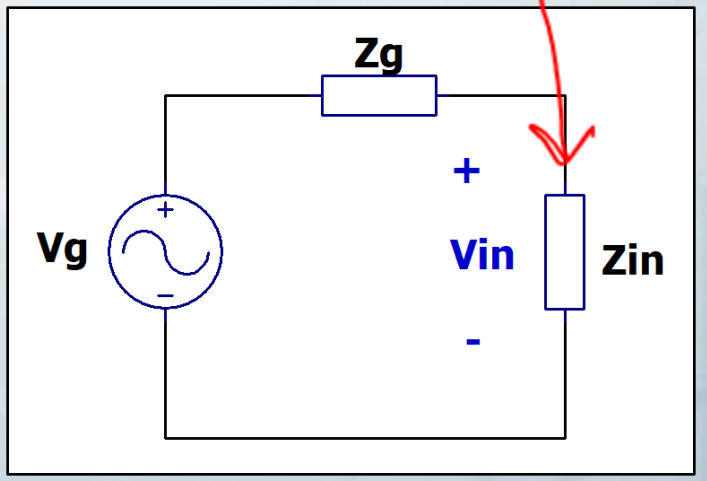
Input Impedance

- Using algebra and trigonometry (Ulaby pg. 72), we can write this as:

$$Z_{in}(z=-L) = Z_0 \frac{Z_L + jZ_0 \tan(\beta L)}{Z_0 + jZ_L \tan(\beta L)}$$

$2\pi/\lambda$ → length of fuet-line



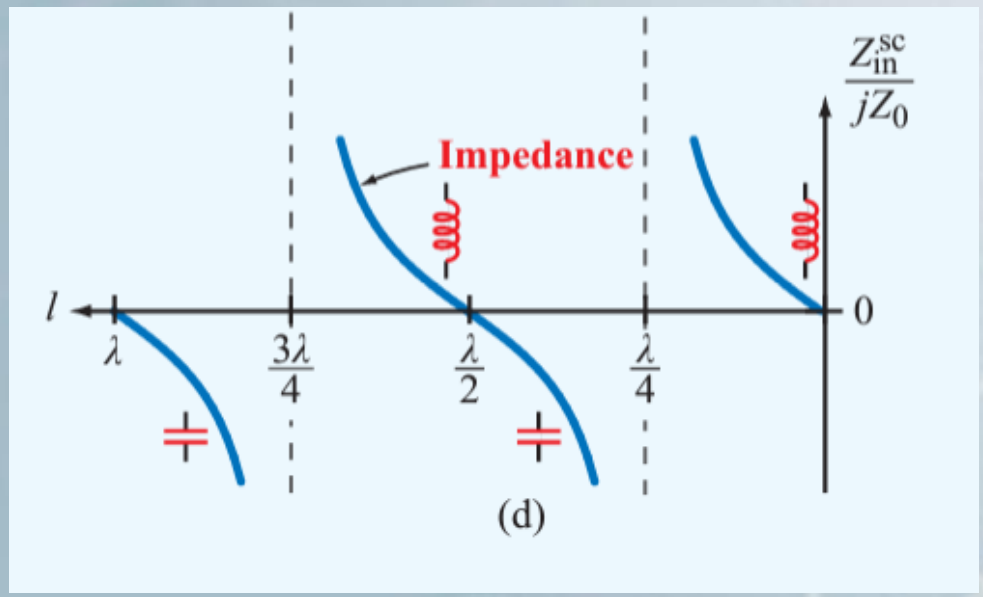


Input Impedance

Special Case: $Z_L=0$ (Short Circuit)

$$Z_{in}(z=-L) = Z_0 \frac{0 + jZ_0 \tan(\beta L)}{Z_0 + j0 \tan(\beta L)} = jZ_0 \tan(\beta L)$$

- For this case, this impedance is imaginary (reactive) and you can vary the t-line length (and also $\boldsymbol{\beta}$ of the wave) to achieve any imaginary impedance you want!



- When $\frac{Z_{in}}{j} > 0$, impedance at the input of the t-line looks like an **inductor**
- When $\frac{Z_{in}}{j} < 0$, impedance at the input of the t-line looks like a **capacitor**
- We can make the t-line look like **any reactive component** by varying its length

Input Impedance

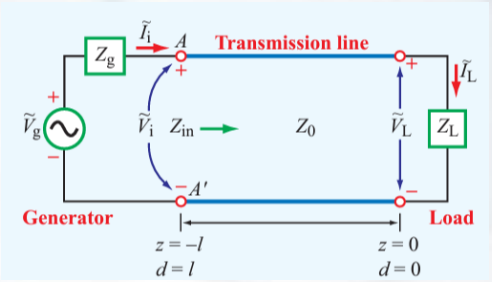
Special Case: $Z_L \to \infty$ (Open Circuit)

$$Z_{in}(z=-L) = Z_0 \frac{\infty + jZ_0 \tan(\beta L)}{Z_0 + j\infty \tan(\beta L)} = -jZ_0 \cot(\beta L)$$

Like the short circuit case, we can make the open-circuited t-line look like **any reactive component** by varying its length

Input Impedance

Special Case: L=0 (No Transmission Line)



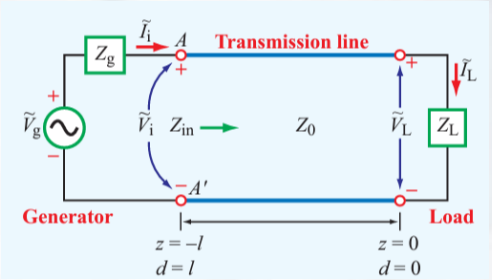
- Recall that the standing wave pattern repeated every half wavelength. We expect that this will also happen for Z_{in}. First, consider the trivial case L=0.

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan(\beta L)}{Z_0 + jZ_L \tan(\beta L)} = Z_0 \frac{Z_L + jZ_0 \tan(0)}{Z_0 + jZ_L \tan(0)} = Z_L$$

If the transmission line has length L = 0, there is **no transmission line** and the **input impedance Z_{in} is that of the load impedance Z_L**

Input Impedance

Special Case: L =$n\lambda/2$ (Half-Wavelength Line)



- Now let the line be a half wavelength long

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan\left(\frac{2\pi}{\lambda}\frac{\lambda}{2}\right)}{Z_0 + jZ_L \tan\left(\frac{2\pi}{\lambda}\frac{\lambda}{2}\right)} = Z_0 \frac{Z_L + jZ_0 \tan(\pi)}{Z_0 + jZ_L \tan(\pi)} = Z_L$$

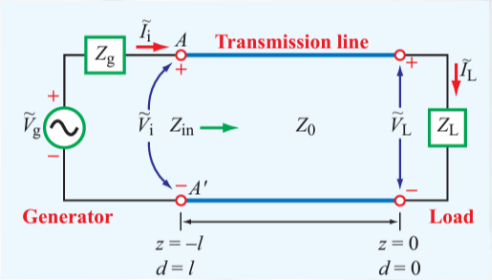
When a transmission line is **half a wavelength long**, **it "looks" as if there is no transmission line there at all**.

The source only "sees" the load impedance even though the t-line's characteristic impedance is not matched to the load impedance ($Z_0 \neq Z_L$). This is due to wave interference effects at the input to the t-line.

$$\frac{\tilde{V}_{in}}{\tilde{I}_{in}} = \frac{\tilde{V}_L}{\tilde{I}_L}$$

Input Impedance

Special Case: L =$n\lambda/2$ (Half-Wavelength Line)



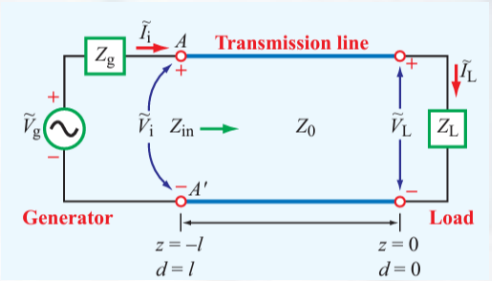
- Thus, for a line that is exactly an integer number of half wavelengths long

$$Z_{in} = Z_L$$

- If you have a transmission line with the wrong characteristic impedance, you can match the load to the source by selecting a length equal to a half wavelength.

Input Impedance

Special Case: L = $n\lambda/4$ (Quarter-Wavelength Line)



- If the line is an odd multiple of a quarter wavelength, we also get an interesting result:

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan\left(\frac{2\pi}{\lambda}\frac{\lambda}{4}\right)}{Z_0 + jZ_L \tan\left(\frac{2\pi}{\lambda}\frac{\lambda}{4}\right)} = Z_0 \frac{Z_L + jZ_0 \tan\left(\frac{\pi}{2}\right)}{Z_0 + jZ_L \tan\left(\frac{\pi}{2}\right)} = Z_0 \frac{Z_0}{Z_L} = \frac{Z_0^2}{Z_L}$$

or

$$\frac{Z_{in}}{Z_0} = \frac{Z_0}{Z_L}$$

Such a transmission line works like an **impedance transformer**. In applications, a quarter-wave t-line can be useful when you want to make one impedance look like another one

Input Impedance

Special Case: L =$n\lambda/4$ Example

$$Z_{in} = \frac{Z_{02}^2}{Z_L}$$

50Ω ↓ 100Ω

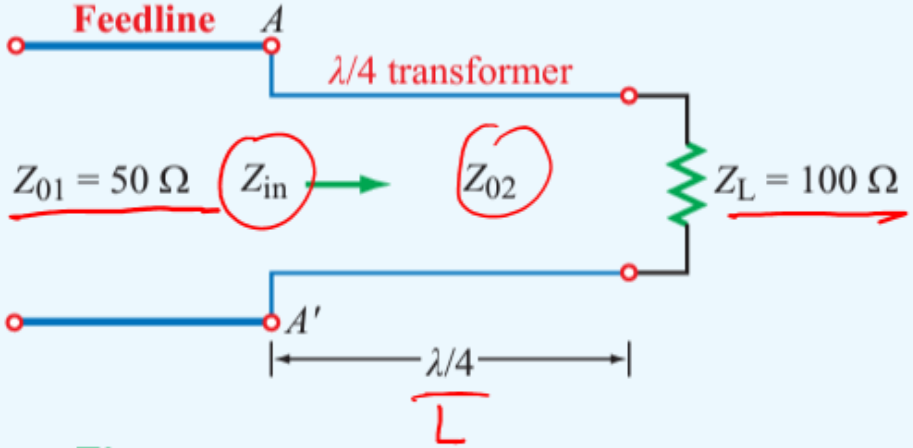


Figure 2-22 Configuration for Example 2-10.

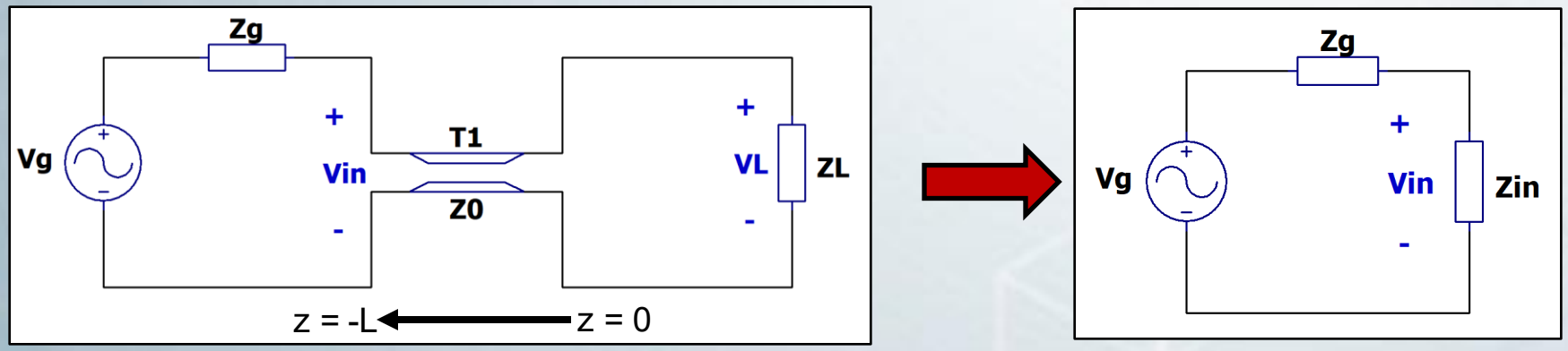
- A lossless transmission line (Z_{01}=50Ω) is to be matched to a resistive load impedance with Z_L=100Ω via a quarter wave section.
- Find the required characteristic impedance Z_{02} of the quarter-wave transformer

for a match between the feedline and the λ/4 transformer : $Z_{in} = 50\Omega$

$$50\Omega = \frac{Z_{02}^2}{100\Omega} \Rightarrow Z_{02} = \sqrt{50\Omega \cdot 100\Omega} = 70.7\Omega$$

Input Impedance

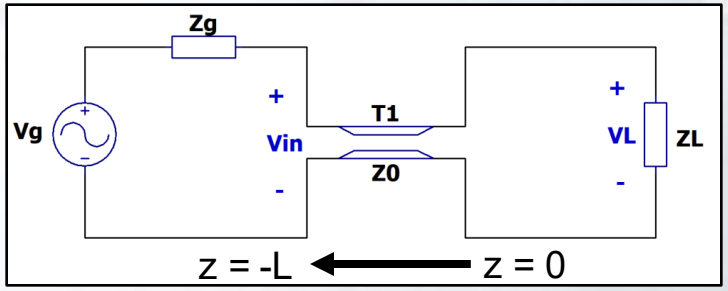
- We've seen that we can use Zin to represent the effect of both the t-line and load in a circuit, but haven't mentioned the voltage source or generator impedance yet.



- So, how do we relate V_g and V_{in} to the t-line voltage wave parameters V_0^+,Γ, β, etc.?

Input Impedance

1. What is $\tilde{V}_{in}(z)$ in terms of the what we know about the voltage on the t-line?



- General expression for t-line voltage

$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}$$

- V_{in} is the voltage at the input to the t-line (z=-L)

$$\tilde{V}_{in} = \tilde{V}(-L) = V_0^+ e^{j\beta L} + V_0^- e^{-j\beta L} = V_0^+ \left(e^{j\beta L} + \Gamma_L e^{-j\beta L}\right)$$

- Solve for V_0^+: we will use this later

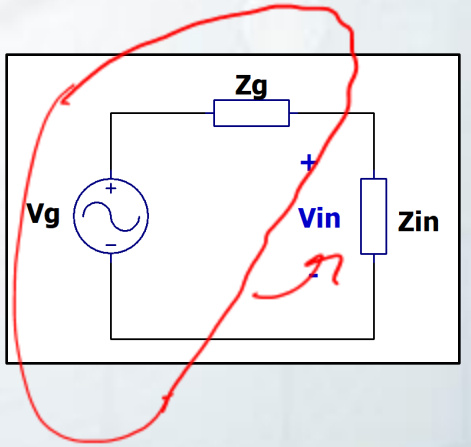
$$V_0^+ = \frac{\tilde{V}_{in}}{e^{j\beta L} + \Gamma_L e^{-j\beta L}}$$

Input Impedance

2. What is $\tilde{V}_{in}(z)$ in terms of the voltage and impedances of the simplified circuit?

   - $\tilde{V}_{in}$ is the voltage divider of $\tilde{V}_g$ between Z_g and Z_{in}:

$$\tilde{V}_{in} = \tilde{V}_g \frac{Z_{in}}{Z_{in} + Z_g}$$



3. Plugging the expression for $\tilde{V}_{in}$ into the expression for V_0^+:

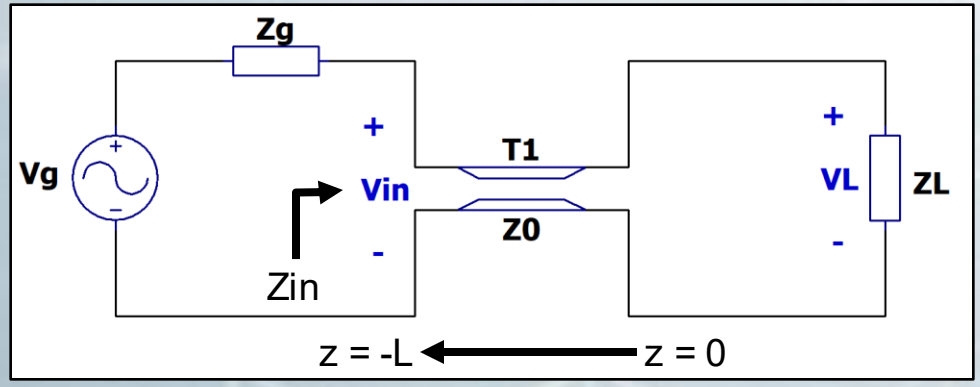
$$V_0^+ = \tilde{V}_g \frac{Z_{in}}{Z_{in} + Z_g} \frac{1}{e^{j\beta L} + \Gamma_L e^{-j\beta L}}$$

Input Impedance

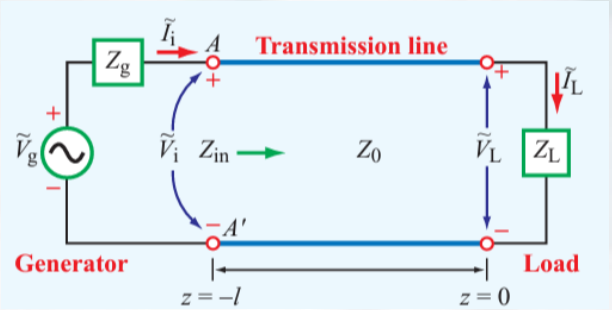
4. Plug the expression for V_0^+ into our general t-line voltage expression $\tilde{V}(z)$ to give the **voltage everywhere on the t-line as a function of V_g and Z_g**:

$$\tilde{V}(z) = \tilde{V}_g \frac{Z_{in}}{Z_{in} + Z_g} \frac{e^{-j\beta z} + \Gamma_L e^{j\beta z}}{e^{j\beta L} + \Gamma_L e^{-j\beta L}}$$

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan(\beta L)}{Z_0 + jZ_L \tan(\beta L)}$$



Input Impedance



- Now, let's look at some special cases to see what this expression predicts

$$\tilde{V}(z) = \tilde{V}_g \frac{Z_{in}}{Z_{in} + Z_g} \frac{e^{-j\beta z} + \Gamma_L e^{j\beta z}}{e^{j\beta L} + \Gamma_L e^{-j\beta L}} \qquad V_0^+ = \tilde{V}_g \frac{Z_{in}}{Z_{in} + Z_g} \frac{1}{e^{j\beta L} + \Gamma_L e^{-j\beta L}}$$

- At the load:

$$\tilde{V}(z = 0) = \tilde{V}_g \left(\frac{Z_{in}}{Z_g + Z_{in}} \right) \frac{1 + \Gamma_L}{e^{j\beta L} + \Gamma_L e^{-j\beta L}}$$

At the load:
- **Γ**=-1 (short circuit) –> V = 0 = Vmin
- **Γ**=+1 (open circuit) –> V = $2V_0^+$ = Vmax
- **Γ**=0 (matched load) –> V = V_0^+: no reflection

Matches T-line theory!

- At the T-line input:

$$\tilde{V}(z = -L) = \tilde{V}_g \left(\frac{Z_{in}}{Z_g + Z_{in}} \right)$$

At t-line input:
- Matches voltage divider between Z_{in} and Z_g
- If L = 0 (no t-line), $Z_{in} = Z_L$ → voltage divider between Z_g and Z_L

Do Lecture 03 Exercise 2 on Gradescope before the beginning of next class.

Wrap-Up

- **No class on Monday (Labor Day)**
- Studio Session 2 next Wednesday